

Shiv Nadar University Chennai

CS3807 – Deep Learning Laboratory

Experiment 7

End-to-End Study of Autoencoders, Convolutional Autoencoders, Denoising Autoencoders and Variational Autoencoders

Degree & Branch	B.Tech Artificial Intelligence & Data Science	Semester V
Subject Code & Name	CS3807 – Deep Learning Laboratory	AY: 2026–27

1. Objective

The objective of this experiment is to develop an end-to-end understanding of autoencoders and their variants for image representation, reconstruction, denoising and generative modeling.

The experiment begins with a fully connected autoencoder, progresses to a Convolutional Autoencoder (CAE), introduces image corruption for denoising, and finally develops a Variational Autoencoder (VAE). Students will visualize the learned latent representation, compare reconstruction quality using quantitative and qualitative measures, and explore the generative capability of the VAE.

The same image dataset and a controlled experimental protocol should be used throughout the experiment wherever possible.

2. Learning Outcomes

After completing this experiment, students will be able to:

- explain the encoder–latent–decoder structure of an autoencoder;
- prepare image data for reconstruction-based learning;
- implement a fully connected autoencoder;
- implement a Convolutional Autoencoder for image reconstruction;
- introduce controlled noise and build a denoising autoencoder;
- evaluate reconstruction using MSE, MAE and SSIM;
- visualize and interpret latent representations;
- explain the probabilistic latent representation of a VAE;
- implement the reparameterization trick;
- calculate and interpret reconstruction and KL-divergence losses;
- generate new images by sampling from a learned latent space; and
- compare deterministic and variational latent representations.

3. Dataset and Experimental Setup

Dataset: MNIST Handwritten Digit Dataset

MNIST contains grayscale images of handwritten digits from 0 to 9. Each image has spatial dimensions

$$28 \times 28 \times 1.$$

The dataset is particularly suitable for this experiment because the images are small, the classes have visually meaningful structure, and the same image format can be directly supplied to fully connected, convolutional and variational autoencoders.

The dataset can be loaded directly using Keras/TensorFlow:

https://www.tensorflow.org/api_docs/python/tf/keras/datasets/mnist

Recommended laboratory subset:

Use 10,000 training images and 2,000 test images for the main laboratory execution. The full MNIST dataset may be used if computational resources permit.

The labels are not required as reconstruction targets. The original image is the target:

$$x \rightarrow \text{Encoder} \rightarrow z \rightarrow \text{Decoder} \rightarrow \hat{x}$$

where x is the input image and $\hat{x}$ is its reconstruction.

4. Expected Input and Output

Expected Input

Each input image has shape

$$28 \times 28 \times 1.$$

For a batch of 32 images:

$$X \in \mathbb{R}^{32 \times 28 \times 28 \times 1}.$$

Pixel values should be converted from

$$[0, 255] \rightarrow [0, 1].$$

For a fully connected autoencoder, the image may be flattened:

$$28 \times 28 = 784.$$

Therefore,

$$x \in \mathbb{R}^{784}.$$

For a convolutional autoencoder, retain the spatial structure:

$$x \in \mathbb{R}^{28 \times 28 \times 1}.$$

Expected Output

The primary output is a reconstructed image:

$$\hat{x} \in \mathbb{R}^{28 \times 28 \times 1}.$$

Students must display:

Original image
Reconstructed image
Reconstruction error

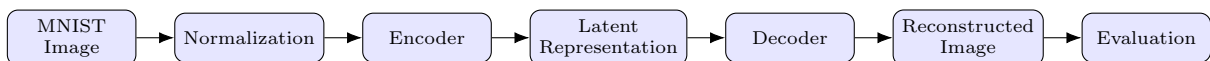
For the denoising model, display:

Original clean image
Noisy image
Denoised reconstruction

For the VAE, display newly generated samples obtained by sampling from the latent distribution.

5. Overall Experimental Pipeline

The complete experiment follows:



The experiment consists of four related studies:

1. Fully Connected Autoencoder
2. Convolutional Autoencoder
3. Denoising Convolutional Autoencoder
4. Variational Autoencoder

6. Exploring Autoencoders

An autoencoder learns to reproduce its input at the output.

The encoder maps the input to a lower-dimensional representation:

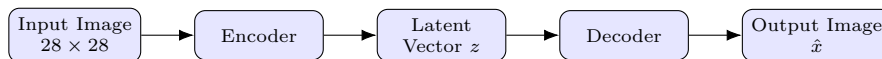
$$z = f_{\theta}(x),$$

and the decoder reconstructs the input:

$$\hat{x} = g_{\phi}(z).$$

The training objective is to minimize reconstruction error:

$$\mathcal{L}_{rec} = \frac{1}{N} \sum_{i=1}^N \|x_i - \hat{x}_i\|^2.$$



The latent vector is a compressed representation of the input.

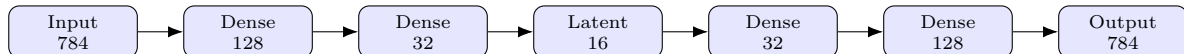
7. Fully Connected Autoencoder

Flatten each image from

$$28 \times 28 = 784$$

to a vector of length 784.

Use the following architecture:



Use ReLU activation in hidden layers and sigmoid activation in the output layer.

Suggested configuration:

Parameter	Value
Latent dimension	16
Optimizer	Adam
Learning rate	10^{-3}
Batch size	128
Epochs	20
Loss	Binary Cross-Entropy

8. Required Analysis of the Basic Autoencoder

Plot 1: Original versus Reconstructed Images

Display at least 10 test images in a grid.

For every image, show:

Original | Reconstruction.

Students must comment on:

- preservation of digit shape;
- loss of fine details;
- blurred regions;
- differences between easy and difficult samples.

Plot 1: Original vs Fully Connected AE Reconstruction

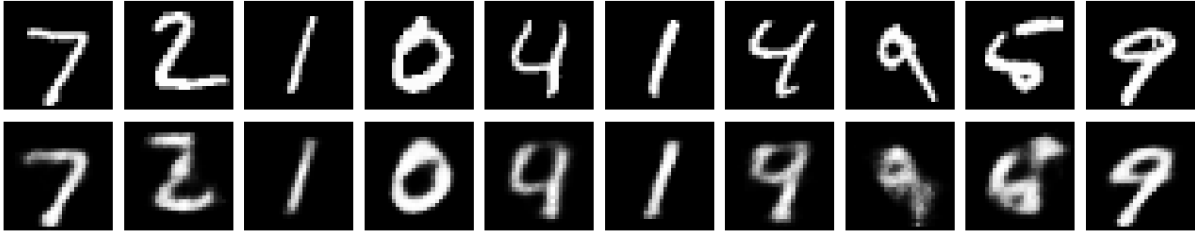


Figure 1: Original and reconstructed images

Inference: we can see that the overall shape of the images is reconstructed but the information about the fine details is gone.

Plot 2: Training and Validation Reconstruction Loss

- X-axis: Epoch
- Y-axis: Reconstruction loss
- Curves: Training loss and validation loss

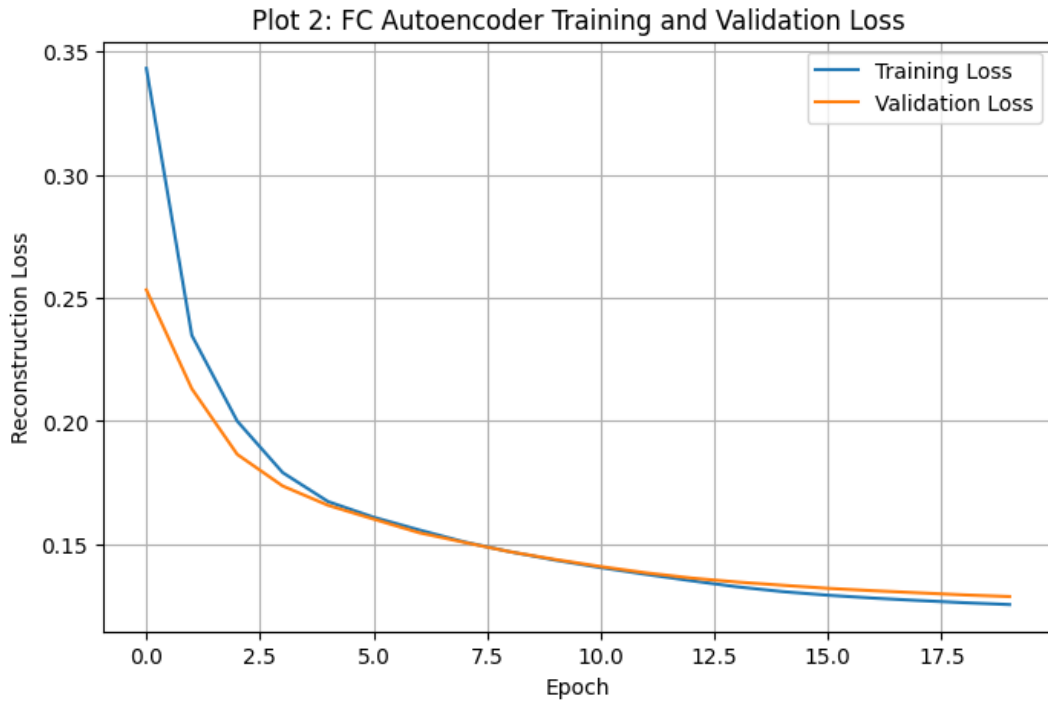


Figure 2: Training and validation reconstruction loss

Inference: we can see that as the autoencoder learns to reconstruct the input images both the losses are decreasing. Similar training and validation trends indicate stable learning.

9. Reconstruction Metrics

Students must calculate the following metrics on the test set.

Mean Squared Error

$$MSE = \frac{1}{N} \sum_{i=1}^N (x_i - \hat{x}_i)^2.$$

Mean Absolute Error

$$MAE = \frac{1}{N} \sum_{i=1}^N |x_i - \hat{x}_i|.$$

Structural Similarity

Calculate SSIM between the original and reconstructed images.

The implementation may use:

<https://scikit-image.org/docs/stable/api/skimetrics.html>

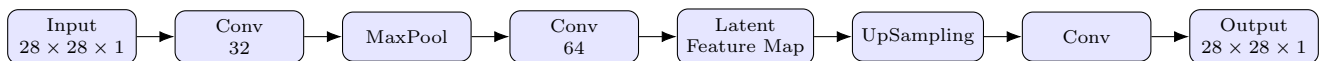
Report:

Metric	Value
Test MSE	
Test MAE	
Mean SSIM	

10. Convolutional Autoencoder

A Convolutional Autoencoder preserves spatial structure and uses convolutional operations in the encoder and upsampling/transposed convolution operations in the decoder.

The conceptual architecture is:



A possible implementation is:

```
Input(28,28,1)
Conv2D(32, 3, activation='relu', padding='same')
MaxPooling2D(2, padding='same')
Conv2D(64, 3, activation='relu', padding='same')
MaxPooling2D(2, padding='same')
Conv2D(64, 3, activation='relu', padding='same')
UpSampling2D(2)
Conv2D(32, 3, activation='relu', padding='same')
UpSampling2D(2)
Conv2D(1, 3, activation='sigmoid', padding='same')
```

The exact implementation may use Conv2DTranspose instead of UpSampling2D.

11. Comparing Fully Connected and Convolutional Autoencoders

Train both models using the same training and test images.

Model	MSE	MAE	SSIM	Parameters
Fully Connected AE				
Convolutional AE				

Plot 3: Reconstruction Comparison

Display:

Original → FC-AE Reconstruction → CAE Reconstruction.

Required inference:

Discuss how preserving spatial locality through convolution affects image reconstruction.

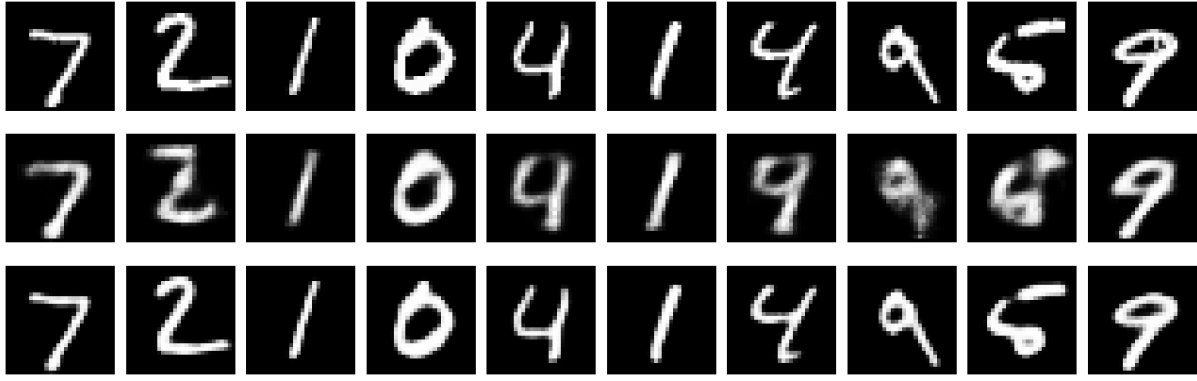


Figure 3: FC-AE and CAE reconstruction comparison

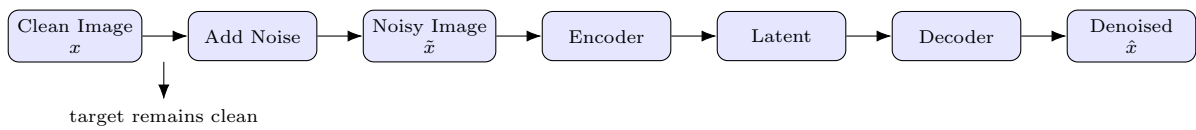
Inference: we can see that convolution layers preserved the spatial information of an image like edges, shapes, and textures. This helps the decoder reconstruct the image more accurately and keeps the important details and structure of the original image.

12. Denoising Autoencoder

A denoising autoencoder receives a corrupted image but is trained to reconstruct the original clean image. The training mapping is:

$$\tilde{x} \rightarrow \text{Encoder} \rightarrow z \rightarrow \text{Decoder} \rightarrow \hat{x}$$

where $\tilde{x}$ is a noisy version of the clean image x .



13. Adding Image Noise

Use at least two corruption types.

Gaussian Noise

$$\tilde{x} = x + n, \quad n \sim \mathcal{N}(0, \sigma^2).$$

Suggested standard deviations:

$$\sigma \in \{0.1, 0.2, 0.3\}.$$

Clip the resulting image to the range $[0, 1]$.

Salt-and-Pepper Noise

Randomly replace a small proportion of pixels with values close to 0 or 1.

Suggested corruption levels:

$$p \in \{0.05, 0.10, 0.20\}.$$

14. Required Denoising Plots

Plot 4: Clean vs. Noisy vs. Denoised

For at least 10 images, display:

Clean | Noisy | Denoised.

Plot 4: Clean vs Noisy vs Denoised

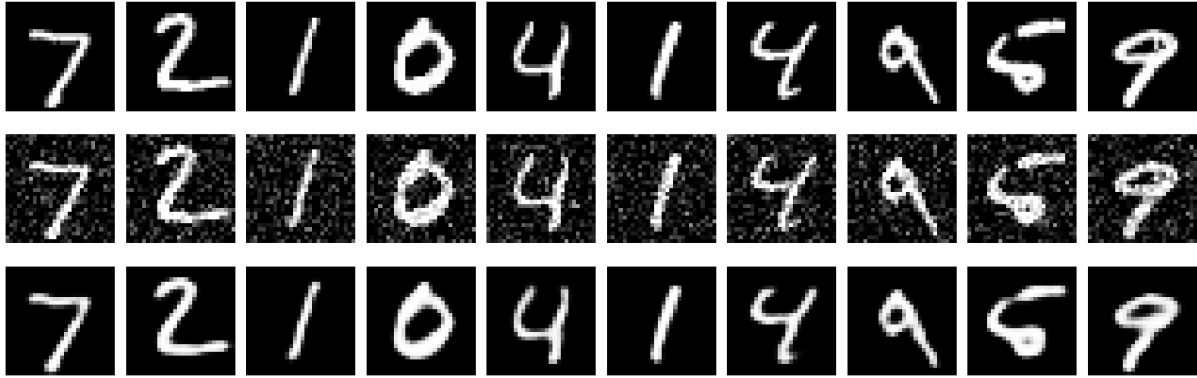


Figure 4: Clean, noisy and denoised images

Inference: we can see that denoising autoencoder has removed most of the noise and recovered the structure of the original digits. **Plot 5: Noise Level vs. Reconstruction Quality**

For each noise level, calculate:

$$MSE, \quad MAE, \quad SSIM.$$

Report:

Noise Level	MSE	MAE	SSIM
0.1			
0.2			
0.3			

Required inference:

Explain how increasing corruption affects reconstruction quality and identify whether the denoising model is able to recover the major structure of the original image.

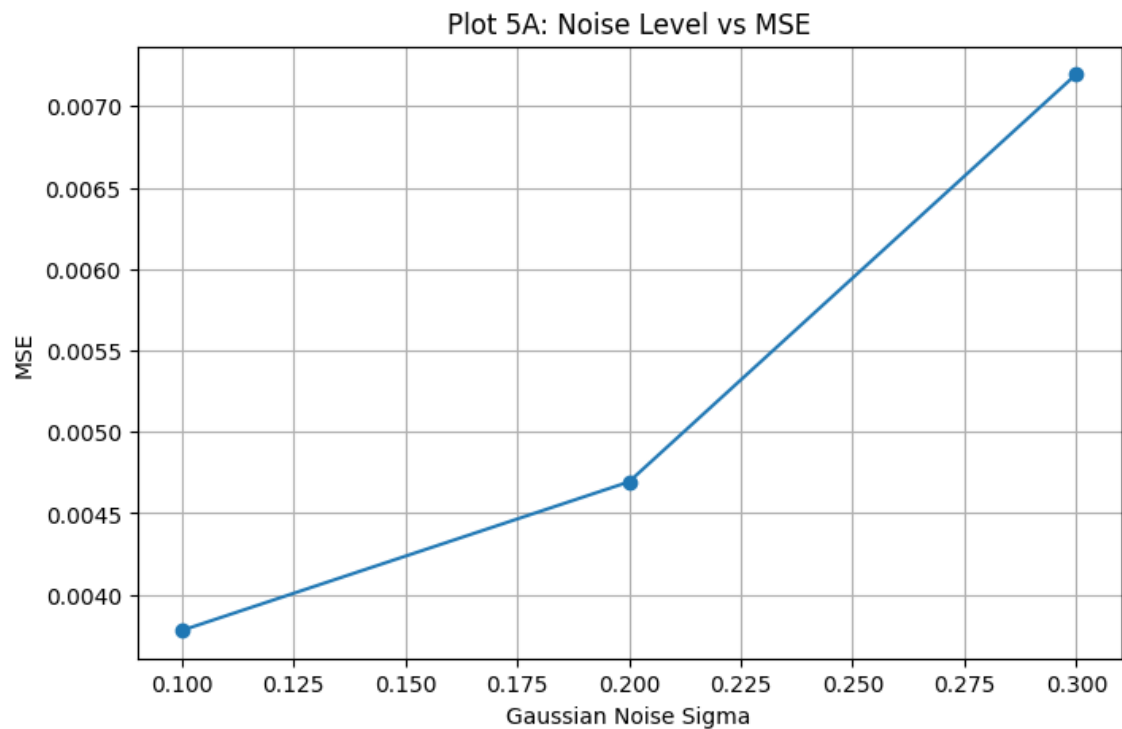


Figure 5: Effect of noise level on reconstruction metrics

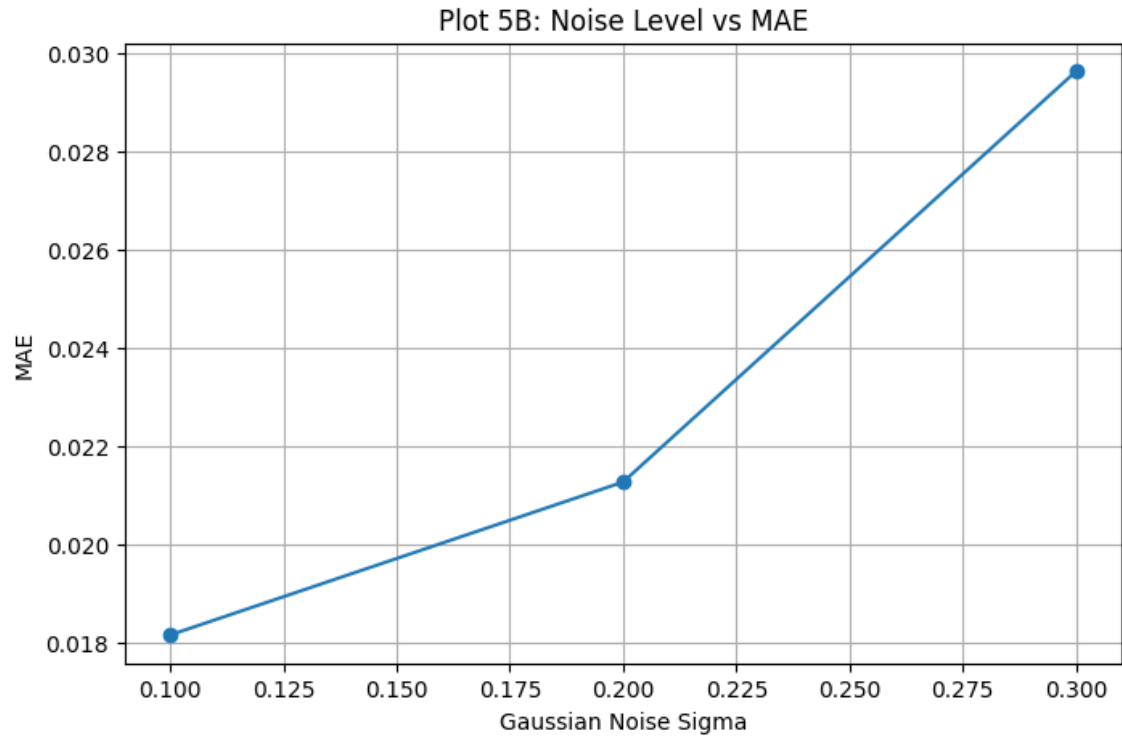


Figure 6: Effect of noise level on reconstruction metrics

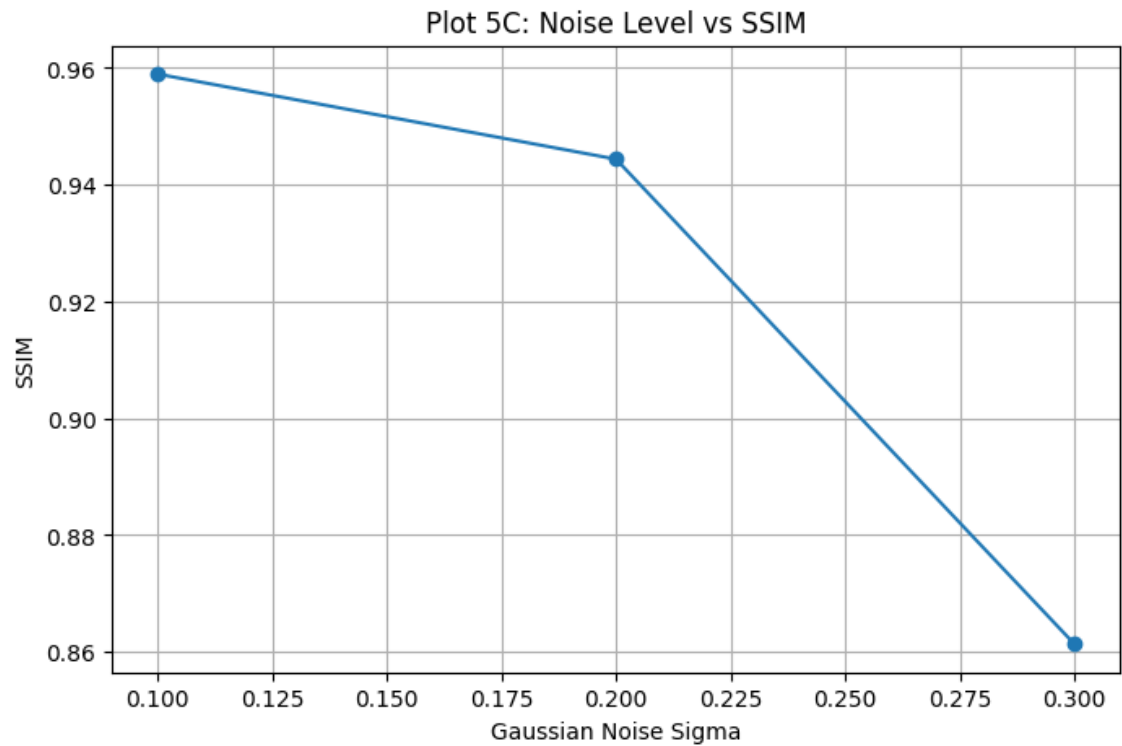


Figure 7: Effect of noise level on reconstruction metrics

Inference: as we can see that the noise level increases, the reconstruction becomes more difficult, so the reconstruction error increases and SSIM decreases. This shows that more noise makes it harder for the model to recover the original image.

15. Variational Autoencoder

A conventional autoencoder maps an input to a deterministic latent vector:

$$z = f(x).$$

A VAE instead learns a probability distribution:

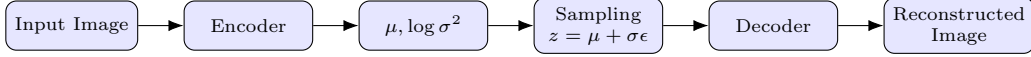
$$q_\phi(z|x) = \mathcal{N}(\mu(x), \text{diag}(\sigma^2(x))).$$

The encoder produces:

$$\mu, \quad \log \sigma^2.$$

A latent vector is sampled using the reparameterization trick:

$$z = \mu + \sigma \odot \epsilon, \quad \epsilon \sim \mathcal{N}(0, I).$$



16. VAE Loss Function

The VAE objective consists of two components.

Reconstruction Loss

$$\mathcal{L}_{rec} = -\mathbb{E}_{q_\phi(z|x)}[\log p_\theta(x|z)].$$

For normalized MNIST images, binary cross-entropy can be used.

KL Divergence

$$D_{KL}(q_\phi(z|x) \parallel p(z)) = -\frac{1}{2} \sum_j (1 + \log \sigma_j^2 - \mu_j^2 - \sigma_j^2).$$

The total loss is

$$\boxed{\mathcal{L}_{VAE} = \mathcal{L}_{rec} + \mathcal{L}_{KL}}$$

where the prior is usually

$$p(z) = \mathcal{N}(0, I).$$

17. VAE Implementation

Use a small latent space so that it can be visualized.

Suggested latent dimension:

$$z \in \mathbb{R}^2.$$

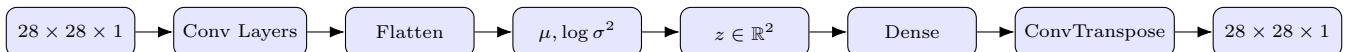
The encoder should produce:

$$\mu \in \mathbb{R}^2, \quad \log \sigma^2 \in \mathbb{R}^2.$$

The decoder maps

$$z \rightarrow 28 \times 28 \times 1.$$

A compact architecture is:



18. Exploring the VAE Latent Space

Since the latent dimension is two, students can visualize the encoded test images in a two-dimensional plane.

Plot 6: VAE Latent Space

- X-axis: z_1
- Y-axis: z_2
- Color/marker category: digit label

The labels are used only for visualization; they are not supplied to the VAE during training.

Required inference:

Students should describe whether samples belonging to the same digit occupy similar regions and whether different digits overlap.

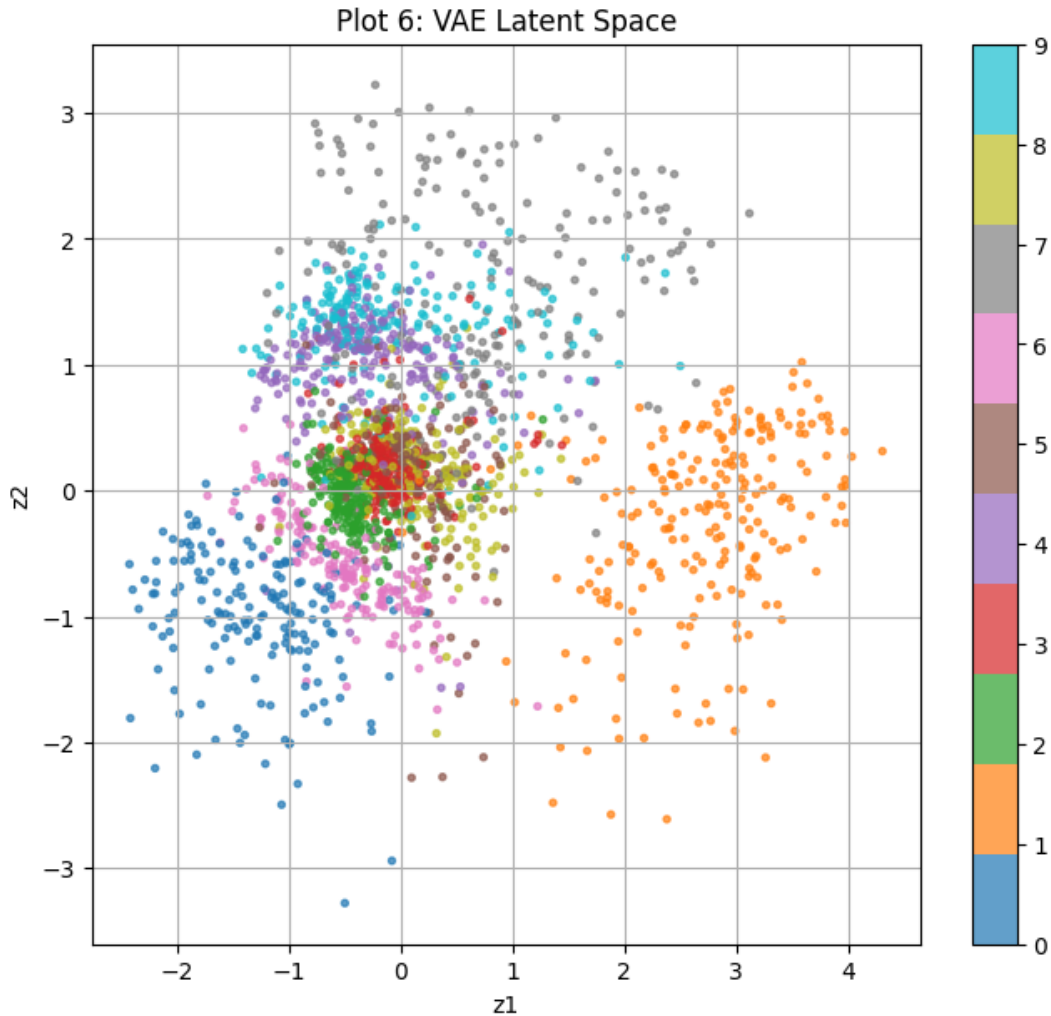


Figure 8: VAE two-dimensional latent space

Inference: samples belonging to the same digit are mostly occupy similar regions of the latent space but some different digits overlap near the center. This shows that the VAE learns some separation between digits but the latent representations are not completely separated.

19. Generating New Images with the VAE

Sample points from

$$z \sim \mathcal{N}(0, I).$$

Pass the sampled latent vectors through the decoder.

Plot 7: Randomly Generated Images

Generate at least 25 images using randomly sampled latent vectors.

Display them in a grid.

Students should determine:

- whether the generated images resemble handwritten digits;
- whether digit shapes are recognizable;
- whether there is visible variation among generated samples; and
- whether some samples are ambiguous or unrealistic.

Plot 7: Randomly Generated VAE Images

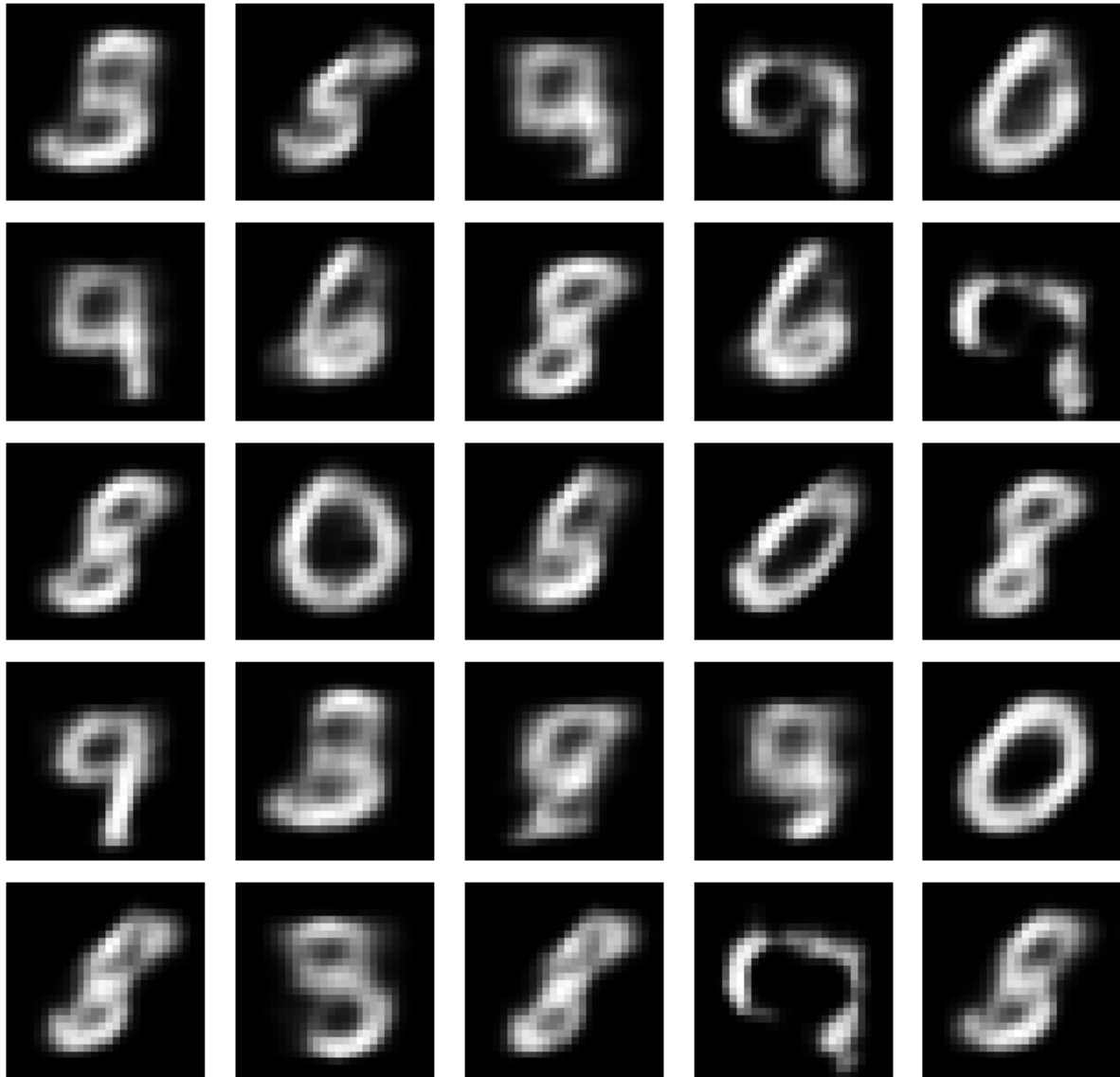


Figure 9: Random images generated by the VAE

Inference: generated images look like handwritten digits. most digits are easy to recognize. There is some changes in the digit shapes. some images are little blur and some digits are not clear and difficult to identify.

20. Latent Space Interpolation

Select two latent vectors:

$$z_A, \quad z_B.$$

Construct

$$z(\alpha) = (1 - \alpha)z_A + \alpha z_B, \quad \alpha \in [0, 1].$$

Use values such as:

$$\alpha = 0, 0.1, 0.2, \dots, 1.$$

Decode every interpolated vector.

Plot 8: Latent Space Interpolation

Display the decoded images in sequence.

Required inference:

Describe whether the decoded image changes smoothly as the latent vector moves between two points.

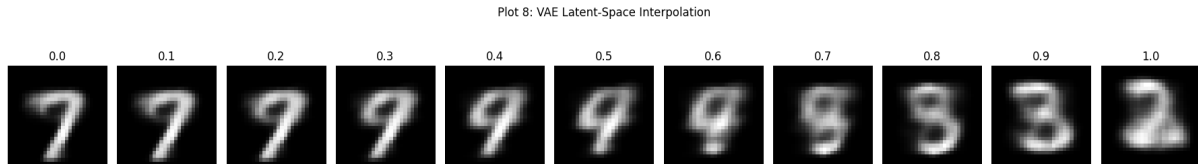


Figure 10: Latent space interpolation

Inference: We can see that the images consecutive to each other are showing gradual and smooth changes.

21. Reconstruction Comparison Across Models

Compare the three reconstruction-based models:

Model	MSE	MAE	SSIM	Parameters
Fully Connected AE	0.022531	0.059488	0.730919	211040
Convolutional AE	0.002523	0.014830	0.974823	74497
Denoising CAE	0.004655	0.021197	0.944980	74497

For the VAE, report its reconstruction loss separately because its total objective also contains the KL-divergence term.

VAE Metric	Value
Reconstruction Loss	148.117371
KL Loss	5.428041
Total Loss	153.545412
Test MSE	0.043924
Test MAE	0.102800
Mean SSIM	0.480729

22. Required Plots

The final report must contain the following plots:

1. Original image versus reconstructed image.
2. Training and validation loss for the basic autoencoder.
3. Fully connected AE versus convolutional AE reconstruction.
4. Clean versus noisy versus denoised images.
5. Noise level versus MSE/MAE/SSIM.
6. VAE latent-space visualization.
7. Randomly generated VAE images.
8. Latent-space interpolation.
9. Training/validation reconstruction loss for the VAE.
10. Reconstruction-error distribution on the test set.

For plots containing multiple quantitative measures, use separate figures or clearly labeled axes rather than placing incompatible scales on one axis.

23. Reconstruction Error Distribution

Calculate the per-image reconstruction error:

$$e_i = \frac{1}{D} \sum_{j=1}^D (x_{ij} - \hat{x}_{ij})^2.$$

Plot 9: Distribution of Reconstruction Errors

Plot a histogram of e_i over the test set.

Students should identify:

- the typical reconstruction-error range;
- whether the distribution is concentrated or spread out;
- possible high-error samples; and
- representative images corresponding to high reconstruction error.

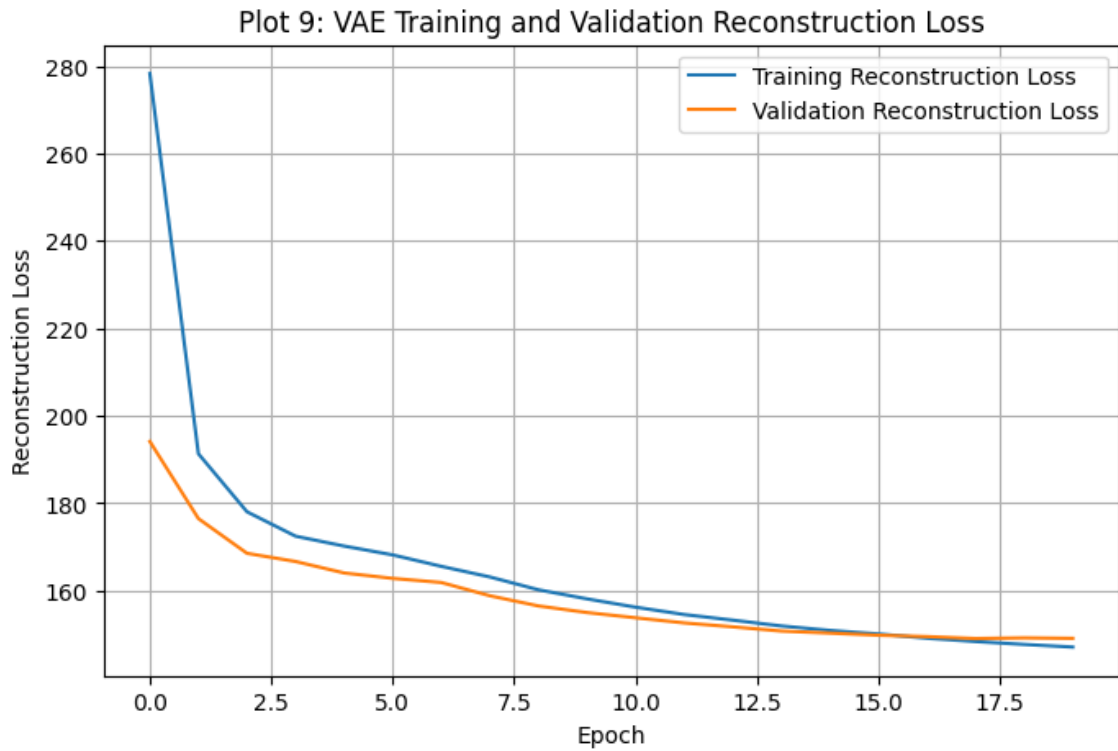


Figure 11: VAE training and validation reconstruction loss

Inference: we can see the as the VAE learns to generate the new images from the distribution of latent space both the losses are decreasing, similar training and validation trends indicate stable learning.

24. Required High-Error Image Analysis

Select the five test images having the largest reconstruction errors.

Display:

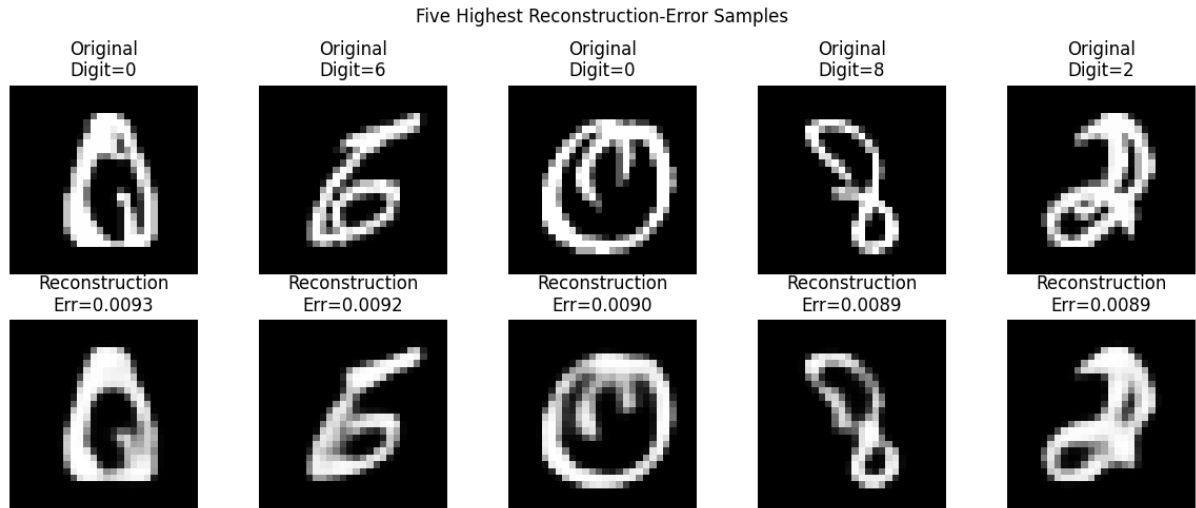


Figure 12: Five images with highest reconstruction error

Inference: highest error samples are generally more difficult to reconstruct because of unusual shapes, ambiguous writing and less defined digit features.

Students should provide a short explanation for why these images may be difficult to reconstruct.

25. Consolidated Results

Model	MSE	MAE	SSIM	Parameters	Time
FC Autoencoder	0.022531	0.059488	0.730919	211040	11.844535
Conv. Autoencoder	0.002523	0.014830	0.974823	74497	19.471153
Denoising CAE	0.004655	0.021197	0.944980	74497	19.788026
VAE	0.043924	0.102800	0.480729	485957	24.105181

The values must be obtained from the student's own execution.

26. Required Inferences

For every major plot, students must write a short inference containing:

1. What does the plot show?
2. What trend is observed?
3. Why might the observed trend occur?
4. What conclusion can be drawn?

The following inferences are mandatory:

- effect of latent dimension on reconstruction;
- difference between fully connected and convolutional reconstruction;
- effect of Gaussian noise level;
- ability of the denoising autoencoder to recover image structure;
- characteristics of the VAE latent distribution;
- structure of the two-dimensional VAE latent space;
- quality and diversity of generated samples;
- smoothness of latent-space interpolation;
- relationship between reconstruction error and visual quality; and
- differences between deterministic and probabilistic latent representations.

27. Additional Latent-Dimension Study

Repeat the basic autoencoder using:

$$d_z \in \{2, 8, 16, 32\}.$$

Report:

Latent Dimension	MSE	SSIM	Parameters
2	0.018742	0.781326	210130
8	0.015284	0.812719	210520
16	0.022531	0.730919	211040
32	0.012821	0.845684	212080

Plot 10: Latent Dimension vs. Reconstruction MSE

Students should explain the trade-off between compression and reconstruction quality.

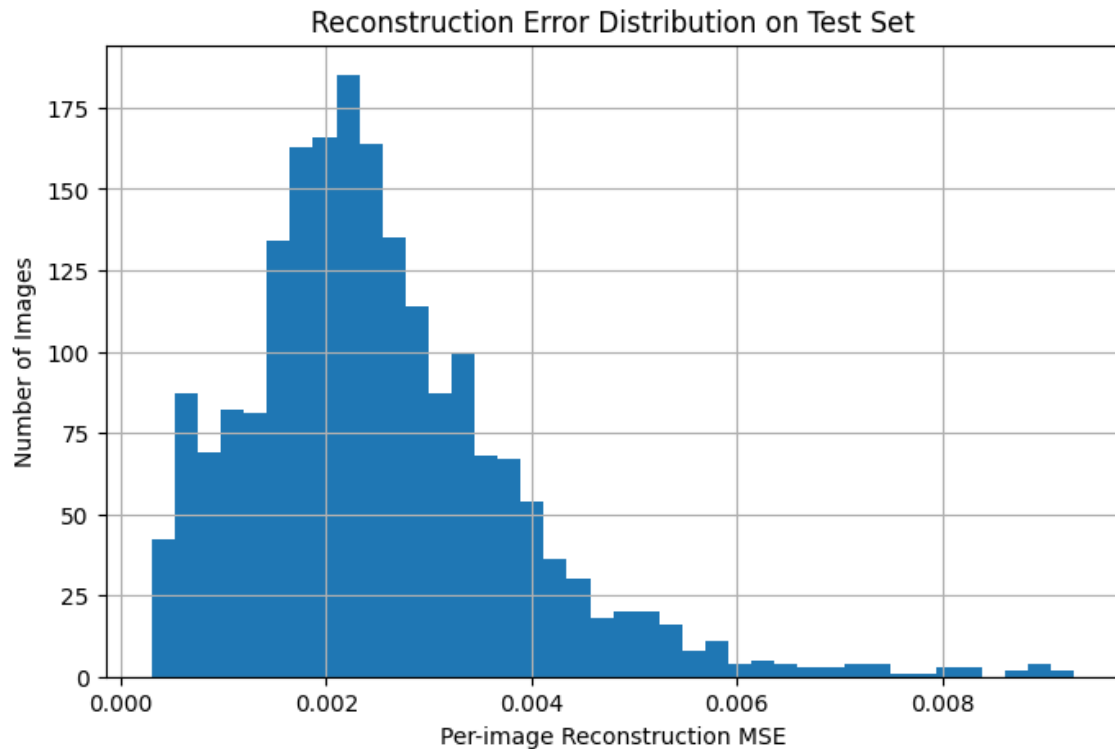


Figure 13: Distribution of reconstruction errors

Inference: most of the test images have reconstruction error in a small range but some images have higher error because the digit shapes are unusual or not clear.

28. Discussion Questions

1. What is an autoencoder?

autoencoder is a neural network which learns to represent input data in a smaller form and then reconstruct the original data from it.

2. What is the purpose of the encoder?

encoder is a neural network that takes the input and converts into small and meaningful representation which is called latent representation.

3. What is the purpose of the decoder? decoder takes the latent representation produced by the encoder and tries to produce the output similar original input.

4. What is a latent representation? latent representation is the form of representation which is compressed and contains the information about the important features needed to represent the data.

5. Why is the latent representation usually smaller than the input? latent representation is smaller because it helps the model to store only the necessary information about important features. This avoids in storing irrelevant information.

6. What happens when the latent dimension is too small? If the latent dimension is small, it may not have the capacity to store more important information, so decoder cannot build the output similar to original output.

7. What happens when the latent dimension is increased? It improves reconstruction because it can store more information, but, it may compress less which results in storing irrelevant information.

8. **Why can a convolutional autoencoder preserve spatial information more effectively?** usually, convolution layers of convolutional autoencoder preserve spatial information such as edges, patterns, shapes.
9. **Differentiate an autoencoder and a denoising autoencoder.** normal autoencoder only learns to reconstruct the original input from the latent representation. denoising autoencoder learns to reconstruct the output similar to original input from noisy input.
10. **Why is the clean image used as the target in denoising?** we try to achieve the clean image. The model learns to denoise to an extent until it reaches clean image.
11. **What happens when the noise level is increased?** more noise level increases difficulty for the model to produce the clean image. This increases reconstruction error and output may not be similar to original input.
12. **Why are MSE, MAE and SSIM useful for image reconstruction?** MSE and MAE is useful because it calculated the difference between the output and the original input which gives us how model is performing. SSIM compares how similar is the output overall structure to the input's.
13. **What is a Variational Autoencoder?** It is a type of auto encoder which learns probability distribution (mean and variance) of latent space and generated a new image with same distribution.
14. **How does a VAE differ from a conventional autoencoder?** conventional autoencoder maps an input to a particular latent representation, but VAE learns the mean and variance of a latent distribution.
15. **Why does a VAE learn a distribution instead of a single latent vector?** It learns distribution because it helps the VAE produce different data with the same distribution and also makes the latent space smoother.
16. **What is the reparameterization trick?** reparameterization trick is a method that allows the VAE to sample a latent vector while still allowing gradients to pass through the sampling process during training.
17. **Why is ϵ sampled from a standard normal distribution?** It gives random noise with a particular distribution which is mixed with the mean and variance that is learned to produce the latent vector.
18. **What is the role of KL divergence in the VAE loss?** KL divergence encourages the learned latent distribution to stay close to the chosen prior distribution.
19. **What is the prior distribution used in the VAE?** standard normal distribution is commonly used in VAE.
20. **Why is a two-dimensional latent space useful for visualization?** two dimensional space is easy to plot in the graph and is easy to observe the formation of the data points.
21. **What does latent-space interpolation demonstrate?** It shows how the generated output changes when we move gradually between two points in the latent space.
22. **Why can a VAE generate new images?** VAE learns the distribution of the latent space. This helps in sampling the new latent points and pass them through the decoder to create new images.
23. **Why might some VAE-generated images be unclear?** some VAE generated images may be unclear because the model generates images by sampling from the latent space, and some sampled points may not contain enough information to produce a clear image.
24. **What is the difference between reconstruction and generation?** Reconstruction is building the output similar to the original input. Generation is creating completely new output from the learned latent representation.
25. **Why should test images remain unseen during training?** we use test images to see how the model performs on new data because model may memorize the training data and perform very good on it.

29. Additional Exercises

1. Train the convolutional autoencoder with latent dimensions of 4, 8, 16 and 32 and compare reconstruction quality.
2. Compare Gaussian noise and salt-and-pepper noise using the same denoising architecture.
3. Train the denoising autoencoder using two different noise levels and compare SSIM.
4. Replace the convolutional decoder with transposed convolutions and compare the reconstructions.
5. Implement a VAE with latent dimension 2 and 8 and compare the latent representations.
6. Investigate the effect of increasing the KL-loss contribution.
7. Generate a larger grid of VAE samples and discuss diversity.
8. Select two digit regions in the latent space and perform interpolation.

30. Expected Outcome

At the end of the experiment, students should be able to demonstrate the complete image-representation and reconstruction pipeline:

Image $\rightarrow$ Encoder $\rightarrow$ Latent Representation $\rightarrow$ Decoder $\rightarrow$ Reconstructed Image

They should also demonstrate denoising:

Clean Image $\rightarrow$ Noise $\rightarrow$ Denoising Autoencoder $\rightarrow$ Recovered Image

and variational generation:

$$\text{Image} \rightarrow (\mu, \sigma) \rightarrow z \rightarrow \text{Decoder} \rightarrow \text{Generated/Reconstructed Image}$$

The final report should contain both quantitative evaluation and visual analysis. Numerical performance values must be obtained from the student's own execution.

31. References

1. Ian Goodfellow, Yoshua Bengio and Aaron Courville, *Deep Learning*, MIT Press, 2016.
2. Diederik P. Kingma and Max Welling, "Auto-Encoding Variational Bayes," ICLR, 2014.
3. Pascal Vincent et al., "Stacked Denoising Autoencoders: Learning Useful Representations in a Deep Network with a Local Denoising Criterion," Journal of Machine Learning Research, 2010.
4. Yann LeCun, Corinna Cortes and Christopher J. C. Burges, "MNIST Handwritten Digit Database."
5. TensorFlow Documentation: <https://www.tensorflow.org>
6. Keras Documentation: <https://keras.io>
7. scikit-image Documentation: <https://scikit-image.org>

GitHub Repository

The complete implementation and source code for this experiment is available on GitHub.

<https://github.com/Nitheesha-1025/Deep-Learning/tree/Autoencoders%20VAE%20CAE>